

On Edge-Balance Index Sets of Regular Graphs

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Abstract

In a bi-racial country, it is desired that the numbers of government ministers from the two races should differ by at most one for fairness. Moreover, the numbers of pairs of ministries which interact directly with each other and both headed by ministers of one race should differ by at most one from that of the other race. One can naturally model the above social phenomenon by a graph labeling which is called edge-balanced, which assigns edges approximately half 0's, and the other half 1's. And then the labeling requires that the induced vertex labels are also 0's over approximately one half vertices, and 1's over the other one half vertices.

We generalize the concept of edge-balanced labeling to the concept of edge-balance index sets of graphs. In this article, properties regarding the edge-balance index sets of regular graphs are investigated. In particular, we completely determine the edge-balance index sets of all 2-regular graphs, Möbius ladders (as examples of 3-regular graphs), and moreover, complete graphs (as examples of general regular graphs) among others.

Keywords: edge-balanced labeling, edge-friendly labeling, edge-balance index, Möbius ladder, regular graphs.

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1 Introduction and Terminology

In 1995[4], M. Kong and S.-M. Lee considered a new labeling problem of graph theory, which is inspired from the following situation. Let us imagine that in a bi-racial country, it is desired that the numbers of government ministers from the two races should differ by at most one for fairness. Moreover, the numbers of pairs of ministries which interact directly with each other and both headed by ministers of one race should differ by at most one from that of the other race. We can naturally model this situation by graph labeling. Each vertex represents a ministry. Two vertices are joined by an edge if and only if the ministries they represent interact directly with each other. We seek a partition of the vertices into two sets which satisfies certain condition of balance, and may define the following concepts.

Definition 1.1 An edge labeling $f : E(G) \rightarrow \{0, 1\}$ of the graph G is said to be **edge-friendly** if $|e_f(0) - e_f(1)| \leq 1$, where $e_f(i)$ is the cardinality of the set $\{e \in E(G) : f(e) = i\}$ for $i \in \{0, 1\}$.

An edge-friendly labeling f of a graph G induces a partially defined vertex labeling f^+ as follows. We define $f^+(v) = 0$, if at the vertex v the number of edges (incident with v) labeled 0 is more than the number of edges labeled 1. Similarly, define $f^+(v) = 1$, if at the vertex v the number of edges labeled 1 is more than the number of edges labeled 0. $f^+(v)$ is not defined otherwise.

Definition 1.2 A graph G is said to be **edge-balanced** if there is an edge-friendly labeling f

satisfying $|v_f(0) - v_f(1)| \leq 1$, where an edge labeling $f : E(G) \rightarrow \{0, 1\}$ is said to be edge-friendly if $|e_f(0) - e_f(1)| \leq 1$, where $v_f(i)$ is the cardinality of the set $\{v \in V(G) : f^+(v) = i\}$, for $i = 0, 1$, and $e_f(i)$ is the cardinality of the set $\{e \in E(G) : f(e) = i\}$ for $i = 0, 1$.

One can see that the edge-balanced labeling describes the above mentioned situation. In [1] the concept of edge-balanced graphs was extended to multigraphs. It was proved that all connected simple graphs except the star $K_{1,2k+1}$, where $k > 0$, are edge-balanced.

In this article only finite simple undirected graphs are considered. We extend the above concept of edge-balanced-ness to a more general situation.

Definition 1.3 The value $|v_f(0) - v_f(1)|$ is called an edge-balance index of G under an edge-friendly labeling f . The edge-balance index set of the graph G , denoted by $EBI(G)$, is defined as the set of all possible edge-balance indices of G , that is the set $\{|v_f(0) - v_f(1)| : f \text{ is an edge friendly labeling of } G\}$.

One can see that if $EBI(G) \subseteq \{0, 1\}$, then the graph is edge-balanced. Hence the notion of edge-balance indices generalizes that of edge-balanced labeling in the sense that if the edge-balance index set for a graph G is known, then the edge-balanced-ness of G is determined.

In order to analyze edge-balance index of a graph G , we may consider the edge induced subgraphs from 0-edges. The calculation of the indices is translated into finding certain types of edge induced subgraphs of G . We define the following:

Definition 1.4 For a graph G with labeling f , let G_0 be a partial subgraph of G with respect to f , which is obtained from deleting all edges labeled 1, and we call G_0 to be the edge induced subgraphs from 0-edges. By $p_k(G)$ we denote the number of vertices of degree k in a graph G , and we use p_k if it is clear from context.

Therefore, if the labeling f is edge-friendly, we have $|E(G_0)| = \lceil \frac{|E(G)|}{2} \rceil$. Since we may change the labeling of 0 and 1 such that $v_f(0) \geq v_f(1)$, therefore without loss of generality we may assume $v_f(0) \geq v_f(1)$ and have another definition of $EBI(G) = \{v_f(0) - v_f(1) \geq 0 : f \text{ is an edge friendly labeling of } G\}$.

Example 1.1 The edge-balance index set of 1-regular graphs nK_2 , $EBI(nK_2)$, is $\{0\}$ if n is even, and is $\{2\}$ if n is odd.

Example 1.2 The edge-balance index set $EBI(St(n))$ of the star graph $St(n)$ with n pendant edges is $\{0\}$ if n is even, and is $\{2\}$ if n is odd.

2 Edge-Balance Index Sets of Paths and Cycles

In the following we calculate EBI for paths, cycles, and union of disjoint cycles (2-regular graphs), using an approach which calculates the edge-balance indices by considering the existence of certain types of edge induced subgraphs. First of all, we need the following lemma.

Lemma 2.1 For any graph G with an labeling f (not necessarily friendly), we have $v_f(0) - v_f(1) = 2v_f(0) + h_f - |V(G)|$, where h_f is the cardinality of the set $\{v \in V(G) : f^+(v) \text{ is not defined}\}$.

Proof.

Since $v_f(0) + v_f(1) + h_f = V(G)$, then we have $v_f(0) - v_f(1) = 2v_f(0) + h_f - (v_f(0) + v_f(1) + h_f) = 2v_f(0) + h_f - |V(G)|$.

Q.E.D.

Lemma 2.2 For any 2-regular graph with an labeling f (not necessarily friendly), we have $v_f(0) - v_f(1) = e_f(0) - e_f(1)$.

Proof. Since each vertex $v \in V(G)$ is of degree 2, it is clear that $f^+(v) = 0$ if and only if 0 is labeled on both edges incident with v , $f^+(v) = 1$ if and only if 1 is labeled on both edges incident with v , and $f^+(v)$ is undefined if and only if 0 is labeled on one incident edge and 1 is labeled on the other.

If we consider the edge induced subgraph G_0 as before, we have $v_f(0) = p_2(G_0)$, $v_f(1) = p_0(G_0)$, $h_f = p_1(G_0)$, and by the degree formula $2p_2(G_0) + p_1(G_0) = 2|E(G_0)| = e_f(0)$. Note that $V(G_0) = E(G_0) = e_f(0) + e_f(1)$, hence $v_f(0) - v_f(1) = 2v_f(0) + h_f - |V(G)| = 2p_2(G_0) + p_1(G_0) - |E(G)| = 2e_f(0) - (e_f(0) + e_f(1)) = e_f(0) - e_f(1)$.

Q.E.D.

The following proposition was shown in [6], and for completion we give an alternative proof using the approach in this article:

Proposition 2.1 For any 2-regular graph G with p vertices, $EBI(G) = \{0\}$ if p is even, and $EBI(G) = \{1\}$ if p is odd.

Proof. Since f is edge-friendly, $v_f(0) - v_f(1) = e_f(0) - e_f(1) = \pm 1$ for n odd, and 0 for n even.

Q.E.D.

Proposition 2.2

$$EBI(P_n) = \begin{cases} \{2\}, & n = 2 \\ \{0\}, & n = 3 \\ \{1, 2\}, & n = 4 \\ \{0, 1\}, & n \geq 5 \text{ is odd} \\ \{0, 1, 2\}, & n \geq 6 \text{ is even} \end{cases}$$

Proof.

The cases for $n = 1, 2, 3, 4$ are easy to check, and for $n \geq 5$, let $P_n = v_1 v_2 \dots v_n$ and let f be an edge friendly labeling of G . Let G_0 be the edge induced subgraph obtained from deleting all edges labeled 1. Since P_n has $n-1$ edges, $E(G_0) = \lceil \frac{n-1}{2} \rceil$ or $\lfloor \frac{n-1}{2} \rfloor$, we observe the following facts for G_0 :

1. for $v = v_1$ or v_n , $f^+(v) = 0$ if $\deg_{G_0}(v) = 1$, $f^+(v) = 1$ if $\deg_{G_0}(v) = 0$.
2. for $v \in V(P_n) - \{v_1, v_n\}$, $f^+(v)$ is undefined when $\deg_{G_0}(v) = 1$, $f^+(v) = 0$ when $\deg_{G_0}(v) = 2$, and $f^+(v) = 1$ whenever $\deg_{G_0}(v) = 0$.

Denote by p_k the number of vertices of degree k in G_0 , for $k = 0, 1, 2$, and $h_f = |\{v \in G : f^+(v) \text{ is undefined}\}| = |\{v \in G_0, \deg(v) = 1\}| = p_1$. By the well known degree formula, we have $2p_2 + p_1 = 2|E(G_0)| = 2\lceil \frac{n-1}{2} \rceil$ or $2\lfloor \frac{n-1}{2} \rfloor$, and with the possible degrees of v_1 and v_n in G_0 , we have the following three cases:

Case 1: $\deg(v_1) = 0, \deg(v_n) = 1$ or $\deg(v_1) = 1, \deg(v_n) = 0$:

we have $h_f = p_1 - 1$, $v_f(0) = p_2 + 1$, $v_f(1) = n - (p_2 + 1) - (p_1 - 1)$, and $|v_f(0) - v_f(1)| = |p_2 + 1 - (p - p_2 - p_1)| = |(2p_2 + p_1) - n + 1| = |2\lceil \frac{n-1}{2} \rceil - n + 1|$ or $|2\lfloor \frac{n-1}{2} \rfloor - n + 1|$.

Case 2: $\deg(v_1) = \deg(v_n) = 1$:

we have $h_f = p_1 - 2$, $v_f(0) = p_2 + 2$, $v_f(1) = n - (p_2 + 2) - (p_1 - 2)$, and $|v_f(0) - v_f(1)| = |p_2 + 2 - (p - p_2 - p_1)| = |(2p_2 + p_1) - n + 2| = |2\lceil \frac{n-1}{2} \rceil - n + 2|$ or $|2\lfloor \frac{n-1}{2} \rfloor - n + 2|$.

Case 3: $\deg(v_1) = \deg(v_n) = 0$:

we have $h_f = p_1$, $v_f(0) = p_2$, $v_f(1) = n - p_2 - p_1$, and $|v_f(0) - v_f(1)| = |p_2 + 2 - (p - p_2 - p_1)| = |(2p_2 + p_1) - n| = |2\lceil \frac{n-1}{2} \rceil - n|$ or $|2\lfloor \frac{n-1}{2} \rfloor - n|$.

Combining the above three cases the proof is complete.

Q.E.D.

In [2], the edge-balance index sets of wheel graphs and fan graphs are determined as given follows:

Example 2.1 For a wheel $W_n = \{v_0\} + C_{n-1}$, let $V(W_n) = \{v_0\} \cup \{v_1, \dots, v_{n-1}\}$ and $E(W_n) = \{v_0 v_i : i = 1, \dots, n-1\} \cup E(C_{n-1})$. Then $EBI(W_n) = \{0, 2, \dots, n-2\}$, if n is even, and $= \{1, 3, \dots, n-2\} \cup \{0, 1, \dots, \frac{(n-1)}{2}\}$, if n is odd.

Example 2.2 For a fan $F_{1,n} = \{v_0\} + P_n$, let $V(F_{1,n}) = \{v_0\} \cup \{v_1, \dots, v_n\}$ and $E(F_{1,n}) = \{v_0 v_i : i = 1, \dots, n\} \cup E(P_n)$. Then

$$EBI(F_{1,n}) = \begin{cases} \{0, 1, 2\}, & n = 3 \\ \{0, 1, \dots, n-2\}, & n \geq 4. \end{cases}$$

3 Edge-Balance Index Sets of General Regular Graphs

In a regular graph G , we have the following observations. If G is odd regular, then G is of even order, and very vertex labeling f^+ induced from an edge friendly labeling f is defined. Also for any $v \in V(G)$, we have $f^+(v) = 0$ if and only if there are at least $\lceil p/2 \rceil$ 0-edges incident with v . Note that we may assume $v_f(0) \geq v_f(1)$. Therefore, the edge-balance index associated with an edge-friendly labeling f of an odd regular graph G is $2v_f(0) - |V(G)|$, and thus every index in $EBI(G)$ is even for odd regular graphs.

The case of even regular graphs is more complicated. In order to analyze the edge-balance indices of regular graphs, we need the following lemmas.

Lemma 3.1 Suppose $nx_n + (n-1)x_{n-1} + \dots + 2x_2 + x_1 = T$, where x_i and T are nonnegative integers, then

1. $x_n + x_{n-1} + \dots + x_m \leq \lfloor \frac{T}{m} \rfloor$
2. $2(x_n + x_{n-1} + \dots + x_m) + x_{m-1} \leq \lfloor \frac{2T}{m} \rfloor$, for $m \geq 2$.

Proof.

Since we have $m(x_n + x_{n-1} + \dots + x_m) \leq \sum_{k=m+1}^n (k-m)x_k + \sum_{i=1}^{m-1} ix_i = T$, therefore, $m(x_n + x_{n-1} + \dots + x_m) \leq T$, and the first statement holds. On the other hand, since $2T = m(2x_n + 2x_{n-1} + \dots + 2x_m + x_{m-1})$, $2\sum_{k=m+1}^n (k-m)x_k + (m-2)x_{m-1} + 2\sum_{i=1}^{m-1} ix_i \geq$

$m(2x_n + 2x_{n-1} + \dots + 2x_m + x_{m-1})$, thus the second statement holds.

Q.E.D.

Lemma 3.2 *Let G be an n -regular (n odd) graph with p vertices, and let f be an edge-friendly labeling of G . Then*

$$\max(EBI(G)) \leq 2\lfloor 2\lceil \frac{n}{2} \rceil^{-1} \lceil \frac{np}{4} \rceil \rfloor - p.$$

Proof.

Let f be a friendly labeling of G . For any $v \in V(G)$, $f^+(v) = 0$ if and only if there are $\lceil n/2 \rceil$ or more 0-edges incident with v . Let G_0 be the edge induced subgraph of G obtained from deleting all the edges labeled by 1. Note that $V(G_0) = V(G)$ and $|E(G_0)| = \frac{|E(G)|}{2} = \lceil \frac{np}{4} \rceil$, and by the degree formula, we have that $\sum_{k=1}^n p_k(G_0) = 2|E(G_0)| = 2\lceil \frac{np}{4} \rceil$. Since $v_f(0) = |\{v \in G_0 : \deg(v) \geq \lceil n/2 \rceil\}| = \sum_{k=\lceil n/2 \rceil}^n p_k(G_0)$, and apply Lemma 3.1, we have that $v_f(0) \leq \lfloor 2\lceil \frac{n}{2} \rceil^{-1} \lceil \frac{np}{4} \rceil \rfloor$, and for any index $r \in EBI(G)$, $r = 2v_f(0) - p \leq 2\lfloor 2\lceil \frac{n}{2} \rceil^{-1} \lceil \frac{np}{4} \rceil \rfloor - p$.

Q.E.D.

Lemma 3.3 *Let G be an n -regular graph (n even) with p vertices, and let f be an edge-friendly labeling of G . Then*

$$\max(EBI(G)) \leq \lfloor 4(\frac{n}{2} + 1)^{-1} \lceil \frac{np}{4} \rceil \rfloor - p.$$

Proof.

Let f be a friendly labeling of G . For any $v \in V(G)$, $f^+(v) = 0$ if and only if there are $\lceil n/2 \rceil$ or more 0-edges incident with v . Let G_0 be the edge induced subgraph of G obtained from deleting all the edges labeled by 1. Note that $V(G_0) = V(G)$ and $|E(G_0)| = \frac{|E(G)|}{2} = \lceil \frac{np}{4} \rceil$, and by the degree formula, we have that $\sum_{k=1}^n p_k(G_0) = 2|E(G_0)| = 2\lceil \frac{np}{4} \rceil$. Since $v_f(0) = |\{v \in G_0 : \deg(v) > \frac{n}{2}\}| = \sum_{k=\frac{n}{2}+1}^n p_k(G_0)$, and $h_f = p_{\frac{n}{2}}(G_0)$, thus we may apply Lemma 3.1, and have that for any index $r \in EBI(G)$, $r = 2v_f(0) + h_f - p \leq \lfloor 4(\frac{n}{2} + 1)^{-1} \lceil \frac{np}{4} \rceil \rfloor - p$.

Q.E.D.

The upper bound in the above lemma is optimal, since we have a lot of examples which achieve the upper bound in this article. The next natural question is that can every number (even) strictly less than the upper bound be realized as an edge-friendly index of a regular graph? For example in

later section we see that Möbius ladders, except the upper bound, have every even number strictly less than the upper bound as an edge-balance index. In fact, we have a more general situation which is proved in [2] for 3-regular graphs:

Lemma 3.4 *Let G be a 3-regular graph with p (even) vertices, and G has a perfect matching. Then for every even number below the upper bound $2\lfloor 3p/4 \rfloor - p$ can be realized as an edge-balance index of G .*

In [2] there are many results about the edge-balance indices of bipartite 3-regular graphs derived from the above key observation, which saves a lot of work while we compute the edge-balance indices for special classes of 3-regular graphs. In particular, the edge-balance index set of bipartite 3-regular graphs with $8k + 4$ and $8k + 6$ vertices is completely determined.

4 Edge-Balance Index Sets of Cubic Graphs

In order to realize the edge-balance indices of more regular graphs, let us define the circulant graph here.

Definition 4.1 *A circulant graph $CIR_n(S)$ is defined by using the vertex set $V(CIR_n(S)) = \{0, 1, 2, \dots, n-1\}$, and the edge set $E(CIR_n(S)) = \{ij | j = i + s, i - s, 1 \leq i \leq n, s \in S\}$, where $S \subset \{1, 2, \dots, \lfloor \frac{n}{2} \rfloor\}$.*

For example, $CIR_n(\{1\}) \cong C_n$, $CIR_n(\{1, 2, \dots, \lfloor \frac{n}{2} \rfloor\}) \cong K_n$, and $CIR_{2n}(\{1, n\}) \cong$ the n -Möbius ladder. We will use the n -Möbius ladders as examples of cubic graphs for which we calculate the edge-balance indices.

An n -Möbius ladder, denoted by $M(n)$, is a graph isomorphic to the circulant graph $CIR_{2n}(\{1, n\})$, which has $2n$ vertices and the i -th vertex is adjacent to the $(i + j)$ -th vertex and the $(i - j)$ -th vertex, for $i = 1, \dots, 2n$ and $j = 1, \dots, n$, where $i + j$ and $i - j$ are calculated modulo $2n$. Note that the Möbius ladder $M(n)$ is cubic with $2n$ vertices and $3n$ edges, and let the vertex set of $M(n)$ be defined as $V(M(n)) = \{v_i : i = 0, 1, \dots, 2n-1\}$ and the edge set of $M(n)$ be $E(M(n)) = \{v_{2n-1}v_0, v_{i-1}v_i, v_jv_{j+n} : i = 1, 2, \dots, 2n-1, j = 0, 1, 2, \dots, n\}$. Also note that $M(n)$ is bipartite if and only if n is odd.

Similarly using the above mentioned key lemmas and constructions whenever necessary, we may calculate all possible edge-balance indices of the Möbius ladder $M(n)$.

Proposition 4.1 *The edge-balance index set $EBI(M(4k+3)) = \{0, 2, \dots, 4k+2\}$, where k is a positive integer.*

Proof.

In this case, $M(4k+3)$ has $8k+6$ vertices. By Lemma 3.2, we have $\max(EBI(M(4k+2))) \leq 2\lceil \frac{3(8k+6)}{4} \rceil - (8k+4) = 4k+4$.

The equality holds when $Y(4k+2)$ has a 2-regular subgraph of order $\frac{3(8k+6)}{4} = 6k+5$. Note that $M(4k+3)$ is a bipartite graph, hence contains no odd cycles and every 2-regular subgraph of $M(4k+3)$ is of even order. Therefore the equality will not hold in this case. With the help of Lemma 3.4, the edge-balance index set $EBI(M(4k+3)) = \{0, 2, \dots, 4k+2\}$.

Q.E.D.

The above proposition gives again an example that the upper bound of $EBI(G)$ in Lemma 3.2 is not always sharp. In the following, we want to find the edge-balance index sets of the other three cases ($n = 4k+2, 4k+1, 4k$) of Möbius ladders $M(n)$.

Proposition 4.2 *The edge-balance index set $EBI(M(4k+2)) = \{0, 2, \dots, 4k+2\}$, where k is a non-negative integer.*

Proof.

In this case, $M(4k+2)$ has $8k+4$ vertices. By Lemma 3.2, $\max(EBI(M(4k+2))) \leq 2\lceil \frac{3(8k+4)}{4} \rceil - (8k+4) = 4k+2$, and by Lemma 3.4, we have $\{0, 2, \dots, 4k\} \subset EBI(M(4k+2))$. It remains to show that $4k+2 \in EBI(M(4k+2))$.

To realize $4k+2$ in $EBI(Y(4k))$, we consider an edge friendly labeling f as following.

$f(v_i v_{i+4k+2}) = f(v_{i-1} v_i) = f(v_i v_{i+1}) = 1$, $i = 1, 5, 9, \dots, 4k+1$, $f(v_j v_{j+4k+2}) = f(v_{j+4k+1} v_{j+4k+2}) = f(v_{j+4k+2} v_{j+4k+3}) = 1$, $j = 3, 7, 11, \dots, 4k-1$, and $f(e) = 0$ for other edges e .

Then $f^+(v) = 1$ for $2k+1$ vertices $v = v_i$, $i = 1, 5, 9, \dots, 4k+1$ and $v = v_{j+4k+2}$, $j = 3, 7, 11, \dots, 4k-1$, and $f^+(v) = 0$ otherwise. Thus $|v_f(0) - v_f(1)| = 2v_f(0) - |V(G)| = 4k+2 \in EBI(M(4k+2))$.

Q.E.D.

Proposition 4.3 *The edge-balance index set $EBI(M(4k+1)) = \{0, 2, \dots, 4k+2\}$, where k is a positive integer.*

Proof.

In this case, $M(4k+1)$ has $8k+2$ vertices. By Lemma 3.2, $\max(EBI(M(4k+1))) \leq 2\lceil \frac{3(8k+2)}{4} \rceil - (8k+2) = 4k+2$, and by Lemma 3.4, we have $\{0, 2, \dots, 4k\} \subset EBI(M(4k+1))$. It remains to show that $4k+2 \in EBI(M(4k+1))$.

To realize $4k+2$ in $EBI(M(4k+1))$, we consider an edge friendly labeling f as following.

$f(v_i v_{i+4k+1}) = f(v_{i-1} v_i) = f(v_i v_{i+1}) = 1$, $i = 1, 5, 9, \dots, 4k-3$, $f(v_j v_{j+4k+1}) = f(v_{j+4k} v_{j+4k+1}) = f(v_{j+4k+1} v_{j+4k+2}) = 1$, $j = 3, 7, 11, \dots, 4k-1$, $f(v_{4k} v_{4k+1}) = 1$, and $f(e) = 0$ for other edges e .

Then $f^+(v) = 1$ for $2k$ vertices, where $v = v_i$, $i = 1, 5, 9, \dots, 4k-3$ and $v = v_{j+4k+2}$, $j = 3, 7, 11, \dots, 4k-1$, and $f^+(v) = 0$ otherwise. Thus $|v_f(0) - v_f(1)| = 2v_f(0) - |V(G)| = 4k+2 \in EBI(M(4k+1))$.

Q.E.D.

Proposition 4.4 *The edge-balance index set $EBI(M(4k)) = \{0, 2, \dots, 4k\}$, where k is a positive integer.*

Proof.

In this case, $M(4k)$ has $8k$ vertices. By Lemma 3.2, $\max(EBI(M(4k))) \leq 2\lceil \frac{3(8k)}{4} \rceil - (8k) = 4k$, and by Lemma 3.4, we have $\{0, 2, \dots, 4k-2\} \subset EBI(M(4k))$. It remains to show that $4k \in EBI(M(4k))$.

To realize $4k$ in $EBI(M(4k))$, we consider an edge friendly labeling f as following.

$f(v_{i-1} v_i) = 0$, $i = 1, 2, \dots, 3k-1$, $f(v_{j-1} v_j) = 0$, $j = 4k+1, 4k+2, \dots, 7k-1$, and $f(v_{4k} v_0) = f(v_{3k-1} v_{7k-1}) = 0$, $f(e) = 1$ for other edges e .

Then $f^+(v) = 0$ for $6k$ vertices, where $v = v_i$, $i = 0, 1, \dots, 3k-1$ and $v = v_j$, $j = 4k, 4k+1, \dots, 7k-1$. Thus $|v_f(0) - v_f(1)| = 2v_f(0) - |V(G)| = 4k \in EBI(M(4k))$.

Q.E.D.

Therefore we may conclude that

Theorem 4.1 *For $n \geq 3$,*

$$EBI(M(n)) = \begin{cases} \{0, 2, \dots, n\}, & n \equiv 0 \pmod{4} \\ \{0, 2, \dots, n+1\}, & n \equiv 1 \pmod{4} \\ \{0, 2, \dots, n\}, & n \equiv 2 \pmod{4} \\ \{0, 2, \dots, n-1\}, & n \equiv 3 \pmod{4} \end{cases}$$

5 Edge-Balance Index Sets of Complete Graphs

We determine the edge-balance index sets of complete graphs K_n in this section. Basic techniques used here are from previous sections.

Theorem 5.1 *For $n = 4k$ or $n = 4k + 2$, $EBI(K_n) = \{0, 2, \dots, n - 2\}$, where k is a positive integer.*

Proof.

Note that K_{4k} is $(4k - 1)$ -regular, and has $2k(4k - 1)$ edges. By Lemma 3.2, we have $4k - 2$ as the upper bound of $EBI(K_{4k})$. To realize $4k - 2$ in $EBI(K_{4k})$, we consider a subgraph of K_{4k} as follows:

$$G_0 \cong CIR_{4k-1}(\{1, 2, \dots, k\}) \cup \{v\}$$

Therefore, $|E(G_0)| = \frac{|E(K_{4k})|}{2} = k(4k - 1)$ and G_0 is $2k$ -regular. Let f be the labeling such that $f(e) = 0$ if and only if $e \in G_0$ and $f(e) = 1$ otherwise. Then $f^+(v) = 0$ for $4k - 1$ vertices under the labeling f , and the upper bound is attained.

As for $0 \leq 2t \leq 4k - 2$, we realize each $2t$ as an edge-balance index using the circulant graphs as follows. Consider a subgraph of K_{4k} using graphs as follows: $CIR_{4k-1}(\{2, \dots, k\}) \cup \{v\}$ with $(k - 1)(4k - 1)$ edges, a path $P_{2k+t} = \{0, 1, 2, \dots, 2k + t\}$ with $(2k + t)$ edges, and $E_t = \{1v, 2v, \dots, (2k + t - 1)v\}$ with $2k + t - 1$ edges. Let $G_{2t} = CIR_{4k-1}(\{2, \dots, k\}) \cup \{v\} \cup P_{2k+t} \cup E_t$, then $|E(G_{2t})| = k(4k - 1) = \frac{|E(K_{4k})|}{2}$. Let f_t be the labeling such that $f_t(e) = 0$ if and only if $e \in G_0$ and $f_t(e) = 1$ otherwise. Therefore $f_t^+(v) = 0$ for $2k + t$ vertices under the labeling f , and the index $r = 2v_f(0) - |V(G)| = 4k + 2t - 4k = 2t$ is realized. The case for $n = 4k + 2$ is similarly obtained.

Q.E.D.

Using the same method, one may have also the following results:

Theorem 5.2 *For $n = 4k + 1$ or $n = 4k + 3$, $EBI(K_n) = \{0, 1, \dots, n - 1\}$, where k is a positive integer.*

6 Concluding Remarks

The edge-balance index sets of regular graphs is discussed in this paper. One may keep working on the edge-balance index sets over various graph

classes, and explore more about their applications to certain balance situations in real life.

On the other hand, the information of edge-balance indices for non-bipartite cubic graphs would be pretty interesting to know and will be helpful to the study of edge-balance indices of general cubic graphs.

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